

Notes: Song, Storesletten, and Zilibotti (AER, 2011)

Growing Like China

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1 What is the paper doing?

China in the 1990s–2000s combines: (i) *very high investment and growth* and (ii) a *large and rising external surplus* (e.g., foreign reserves). In a standard neoclassical open-economy logic, high returns to capital should attract capital inflows and reduce the surplus. The paper proposes a mechanism that *jointly* produces:

- a sustained **reallocation** from less productive but well-financed state firms to more productive but financially constrained private firms,
- **TFP growth** driven by this reallocation,
- a **foreign surplus** because domestic savings cannot be fully absorbed by domestic investment while the productive sector remains credit constrained,
- and **rising inequality** as entrepreneurial rents expand.

Key modeling mapping:

- **F-firms** (“financially integrated”): interpreted as SOEs, have (cheap) access to bank loans, but have lower productivity.
- **E-firms** (“entrepreneurial”): interpreted as private firms, higher productivity, but face borrowing constraints (limited pledgeability).

2 Motivating facts: figures and Table 1

2.1 Fig. 1: Foreign reserves and the deposit–loan gap

Interpretation (intuition). The deposit–loan gap tracks the run-up in foreign reserves: domestic banks collect large deposits but do not expand domestic lending one-for-one, so the residual is invested in foreign assets. This is consistent with an economy where (i) households save a lot, but (ii) the set of domestic borrowers able/allowed to absorb funds is limited.

2.2 Fig. 2: Employment reallocation from SOEs to private firms

Interpretation. The privatization/marketization transition is visible in labor allocation: the share of employment in domestic private enterprises (and, more broadly, non-state firms) rises sharply from the early 1990s to the mid-2000s. This reallocation is the paper’s central “state → private” transition object.

2.3 Fig. 3: Profits over (net) fixed assets by ownership

Interpretation. Private firms are more profitable on average (consistent with higher productivity and higher marginal returns). A key additional fact: SOE profitability does not collapse—it is stable or rising—which motivates a *two-sector + market power* extension later in the paper (SOEs retreat to capital-intensive/strategic sectors where they may earn rents/markups).

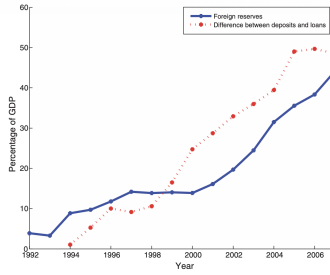


FIGURE 1. FOREIGN RESERVES AND THE DIFFERENCE BETWEEN DEPOSITS AND LOANS

Note: The figure plots China's foreign reserves (solid line) and the domestic bank deposits minus domestic loans (dotted line), both expressed as a percentage of GDP.
Source: CSY, various issues.

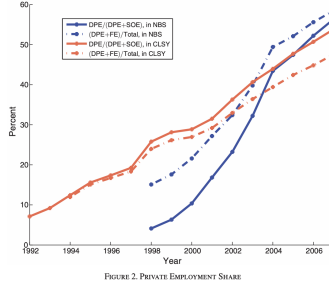


FIGURE 2. PRIVATE EMPLOYMENT SHARE

Note: The figure shows, first, the DPE share of employment as a share of SOE + DPE employment in manufacturing (NBS 1998-2007) and in the urban sector (CLSY 1992-2007). Second, it plots DPE + FE employment as a share of total employment in manufacturing (NBS 1998-2007) and in the urban sector (CLSY 1992-2007).
Source: CSY and CLSY, various issues.

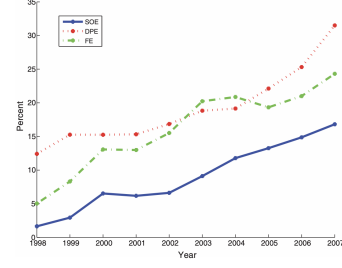


FIGURE 3. TOTAL PROFITS OVER NET VALUE OF FIXED ASSETS

Note: The figure plots the average ratio between total profits and the book value of fixed assets across firms of different ownership, in percent.
Source: CSY, various issues.

2.4 Fig. 4: External finance shares across ownership

Interpretation. SOEs rely much more on formal external finance (bank loans and government budgets) than private firms. Private firms finance investment mainly through retained earnings / internal funds. This is direct reduced-form evidence of **credit discrimination** / imperfect financial access that is central to the model.

2.5 Fig. 5: SOEs concentrate in capital-intensive industries

Interpretation. SOEs are disproportionately present in more capital-intensive industries, and their employment share declines more in less capital-intensive industries. This motivates the paper's **two-sector extension**: credit-favored but less productive SOEs have comparative advantage in capital-intensive production, while productive but constrained private firms expand first in labor-intensive sectors.

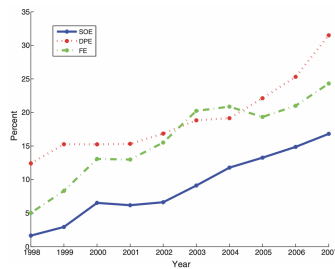


FIGURE 3. TOTAL PROFITS OVER NET VALUE OF FIXED ASSETS

Note: The figure plots the average ratio between total profits and the book value of fixed assets across firms of different ownership, in percent.
Source: CSY, various issues.

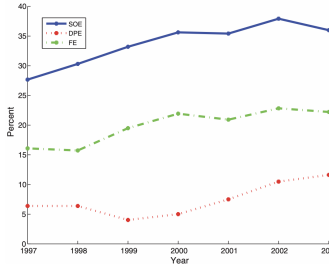


FIGURE 4. SHARE OF INVESTMENT FINANCED BY BANK LOANS AND GOVERNMENT BUDGETS

Note: The figure plots the average share of investment financed by bank loans and government subsidies across firms of different ownership, in percent.
Sources: CSY 1998 to 2001 and 2003, China Economy and Trade Statistical Yearbook 2002 and 2004.

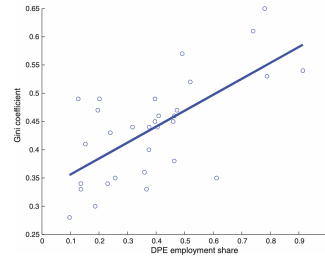


FIGURE 6. INCOME INEQUALITY AND PRIVATE EMPLOYMENT SHARE ACROSS PROVINCES

2.6 Fig. 6: Inequality and the private employment share

Interpretation. The transition comes with rising dispersion in incomes/consumption. In the model, inequality rises because entrepreneurs earn rents and accumulate wealth while workers save at low deposit rates.

2.7 Table 1: Privatization, the external surplus, and growth (reduced form)

Table 1: The private employment share and: external surplus and growth

	$(S - I)/GDP$		GDP p.c. growth		VA per worker growth	
	(1)	(2)	(3)	(4)	(5)	(6)
$D.(EMPL^{PRIV})$	0.9964** (0.4889)	0.8920* (0.4659)	0.1893*** (0.0603)	0.1903*** (0.0610)		
$D.(EMPL^{NONSOE})$					1.4257*** (0.4785)	1.5973*** (0.3572)
$L.(GDP\ p.c.)$		6.6268*** (2.3952)		-0.0646 (0.2136)		
$L.(VA\ p.w.)$						0.1283*** (0.0152)
Year dummy	Yes	Yes	Yes	Yes	Yes	Yes
Observations	124	124	124	124	112	112
R^2	0.0424	0.1984	0.2252	0.2258	0.2104	0.2577

Interpretation / what Table 1 is used for. Table 1 is a reduced-form complement to the model: it documents that where/when the shift toward private firms is faster, both the *external surplus* and *productivity/growth* tend to rise. This is the joint fact the model is designed to rationalize.

3 Benchmark model (two-period OLG): environment and objects

3.1 Demographics and preferences

Time is discrete. A cohort is born each period. Population grows at rate ν :

$$N_{t+1} = (1 + \nu)N_t.$$

Each agent lives two periods: young at t and old at $t + 1$. Preferences (eq. (1)):

$$u(c_t, c_{t+1}) = \frac{c_t^{1-1/\theta} - 1}{1 - 1/\theta} + \beta \frac{c_{t+1}^{1-1/\theta} - 1}{1 - 1/\theta}. \quad (1)$$

θ is the intertemporal elasticity of substitution ($\theta = 1$ gives log utility). A fraction μ of the young are entrepreneurs/managers; the rest are workers.

3.2 Workers: budget constraint and saving rate

A young worker earns wage income w_t , saves s_t^W in deposits, and consumes:

$$c_t^W = w_t - s_t^W, \quad c_{t+1}^W = R s_t^W.$$

With CRRA utility as in (1), optimal saving is linear in income:

$$s_t^W = \zeta^W w_t, \quad \zeta^W = \left(1 + \beta^{-\theta} R^{1-\theta}\right)^{-1}. \quad (2)$$

Interpretation. Workers supply deposits to the banking system; this deposit supply is a key driver of the deposit–loan gap and foreign asset accumulation during transition.

3.3 Production: two firm types (F and E)

Aggregate productivity grows exogenously:

$$A_{t+1} = (1 + z)A_t.$$

There are two technologies (Cobb–Douglas, with different efficiency):

$$y_t^F = k_{F,t}^\alpha (A_t n_{F,t})^{1-\alpha}, \quad (3)$$

$$y_t^E = k_{E,t}^\alpha (\chi A_t n_{E,t})^{1-\alpha}, \quad (4)$$

where $\chi > 1$ captures the TFP advantage of entrepreneurial/private firms.

3.4 Financial sector and interest rates

Banks take deposits from workers and lend to firms. Deposits pay the gross interest rate R :

$$R^d = R.$$

Intermediation has iceberg cost $\xi \in (0, 1)$ so loans must satisfy:

$$R^l = \frac{R}{1 - \xi}.$$

(Interpretation: financial repression/intermediation frictions raise the effective lending rate relative to the deposit rate.)

3.5 Agency and pledgeability

A key friction is limited contractibility and managerial diversion. In E-firms, output is nonverifiable and a manager can divert a share $\psi \in (0, 1)$ unless incentivized. Only a fraction $\eta \in (0, 1)$ of *next-period net profits* can be pledged to outside financiers.

4 Static firm problems and factor prices

4.1 F-firm (SOE-like): competitive, unconstrained borrowing

Given R^l and wage w_t , an F-firm chooses labor to maximize current profit:

$$\max_{n_{F,t}} y_t^F - w_t n_{F,t} - R^l k_{F,t}.$$

Standard Cobb–Douglas FOCs imply a constant capital–effective-labor ratio:

$$\kappa_F \equiv \frac{k_{F,t}}{A_t n_{F,t}} = \left(\frac{\alpha}{R^l} \right)^{\frac{1}{1-\alpha}} \quad (\text{eq. (8) in the paper}). \quad (5)$$

The equilibrium wage is pinned down by the F-firm labor demand:

$$w_t = (1 - \alpha) \left(\frac{\alpha}{R^l} \right)^{\frac{\alpha}{1-\alpha}} A_t \quad (\text{eq. (2) in the paper}). \quad (6)$$

Intuition: during the transition, as long as F-firms operate, wages are determined by their technology and the loan rate R^l .

4.2 E-firm (private): delegation + diversion

An old entrepreneur owns an E-firm with predetermined capital $k_{E,t}$ and hires a young manager. The manager is paid m_t and can otherwise divert a fraction ψ of output. The entrepreneur solves (eq. (3)):

$$\Xi_t(k_{E,t}) = \max_{n_{E,t}, m_t} y_t^E - m_t - w_t n_{E,t} \quad \text{s.t.} \quad m_t \geq \psi y_t^E. \quad (7)$$

The incentive constraint binds, so:

$$m_t = \psi y_t^E \quad (\text{eq. (4)}). \quad (8)$$

Substituting and optimizing over labor yields (eq. (5)):

$$n_{E,t} = ((1 - \psi)\chi)^{1/\alpha} \left(\frac{R^l}{\alpha} \right)^{\frac{1}{1-\alpha}} \frac{k_{E,t}}{\chi A_t}. \quad (9)$$

The resulting value function is linear in $k_{E,t}$ (eq. (6)):

$$\Xi_t(k_{E,t}) = (1 - \psi)^{1/\alpha} \chi^{\frac{1-\alpha}{\alpha}} R^l k_{E,t} \equiv \rho_E k_{E,t}. \quad (10)$$

Assumption 1 (paper's Assumption 1). $\chi > \bar{\chi} \equiv \left(\frac{1}{1-\psi} \right)^{\frac{1}{1-\alpha}}$.

Interpretation of Assumption 1. Despite diversion, E-firms are sufficiently more productive than F-firms so that entrepreneurs prefer the E-technology (delegation) and E-firms can earn higher returns on internal funds.

5 Borrowing constraint, financing mix, and entrepreneurial saving

5.1 Borrowing constraint (limited pledgeability)

Let s_t^E denote savings of a young entrepreneur at t , and ℓ_t^E the bank loan used to finance capital next period:

$$k_{E,t+1} = s_t^E + \ell_t^E.$$

Only a fraction η of next period profits is pledgeable, so the loan contract must satisfy:

$$R^l \ell_t^E \leq \eta \rho_E (s_t^E + \ell_t^E). \quad (11)$$

When the constraint binds (paper assumes parameters such that it does), it implies:

$$\ell_t^E = \frac{\eta \rho_E}{R^l - \eta \rho_E} s_t^E, \quad k_{E,t+1} = \frac{R^l}{R^l - \eta \rho_E} s_t^E.$$

The **share of investment financed by bank loans** is (eq. (7)):

$$\frac{\ell_t^E}{\ell_t^E + s_t^E} = \frac{\eta \rho_E}{R^l}. \quad (12)$$

Interpretation. Even if entrepreneurs have positive-NPV projects, they cannot scale them up quickly because only a limited part of future profits can be pledged. As a result, the private sector grows by *self-financing* (retained earnings).

5.2 Entrepreneurial saving rule

A young entrepreneur earns the manager income m_t and chooses s_t^E to maximize lifetime utility, taking into account that borrowing expands capital one-for-one via the leverage factor $R^l/(R^l - \eta\rho_E)$. The paper shows the optimal saving is linear in income:

$$s_t^E = \zeta^E m_t, \quad \zeta^E \equiv \left(1 + \beta^{-\theta} \left(\frac{(1-\eta)\rho_E R^l}{R^l - \eta\rho_E} \right)^{1-\theta} \right)^{-1}. \quad (13)$$

6 Transition dynamics in the benchmark model

6.1 Capital intensity ordering (Lemma 1)

Define capital per effective labor:

$$\kappa_F \equiv \frac{k_F}{An_F}, \quad \kappa_E \equiv \frac{k_E}{\chi An_E}.$$

Using (9) and (5), the paper derives (eq. (9)):

$$\kappa_E = \kappa_F ((1-\psi)\chi)^{-1/\alpha}. \quad (14)$$

Lemma 1 (Lemma 1). *During the transition (when both firm types operate), E-firms have lower capital-output and capital-labor ratios than F-firms.*

Intuition. Because E-firms are financially constrained and pay an effective “tax” through diversion, they behave as if capital is relatively expensive; they are therefore more labor intensive.

6.2 Entrepreneurial-capital growth (Lemma 2)

The key object is the growth rate of entrepreneurial capital. First note that (using Cobb–Douglas shares and the binding incentive constraint) the manager income is proportional to capital:

$$m_t = \psi y_t^E = \frac{\psi}{1-\psi} \frac{\rho_E}{\alpha} k_{E,t}.$$

Then, combining the saving rule (13) with the leverage mapping $k_{E,t+1} = \frac{R^l}{R^l - \eta\rho_E} s_t^E$ yields the growth factor:

$$1 + \gamma_{K_E} = \frac{R^l}{R^l - \eta\rho_E} \zeta^E \frac{\psi}{1-\psi} \frac{\rho_E}{\alpha} \quad (\text{eq. (10)}). \quad (15)$$

Lemma 2 (Lemma 2). *There exists a threshold $\hat{\chi}$ such that if $\chi > \hat{\chi}$, the E-sector expands over time (its employment share rises). Moreover, $\hat{\chi}$ is decreasing in β and η , and increasing in ν and z .*

Interpretation. Higher productivity advantage χ , better pledgeability η , and greater patience β speed up private-sector capital accumulation and hence the transition.

6.3 Aggregate output per worker

Let $N_{E,t}$ denote workers employed by E-firms and N_t total workers. The paper shows (eq. (11)):

$$\frac{Y_t}{N_t} = \kappa_F^\alpha \left(1 + \frac{\psi}{1-\psi} \frac{N_{E,t}}{N_t} \right) A_t. \quad (16)$$

Interpretation. Output per worker grows both because A_t grows exogenously and because the share of labor employed by E-firms rises endogenously (reallocation-driven TFP growth).

6.4 Fig. 7: Transition dynamics (analytical model)

How we read Fig. 7. As E-firms accumulate capital via retained earnings, their employment share rises. Since E-firms are more productive, aggregate output and measured TFP rise. Because E-firms are constrained (cannot absorb deposits quickly), banks accumulate foreign assets, generating a sustained external surplus during transition.

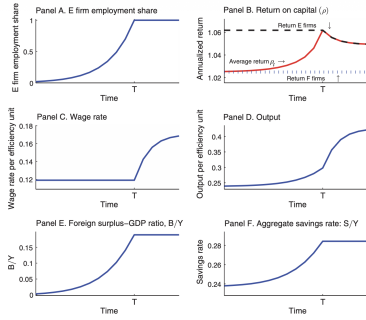


FIGURE 7. TRANSITION IN THE ANALYTICAL MODEL.
Notes: The figure shows the evolution of key variables during and after the transition in the analytical model. Time T denotes the end of the transition, when all workers are employed in E-firms.

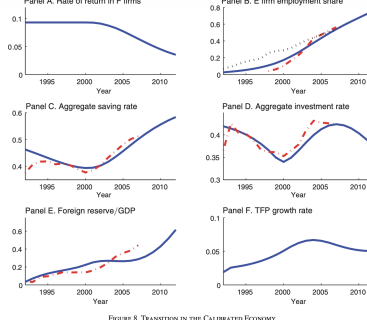


FIGURE 8. TRANSITION IN THE CALIBRATED ECONOMY.
Notes: The figure shows the evolution of key variables during and after the transition in the calibrated economy. The solid and dashed lines refer to the simulated results from the model and the data, respectively. The dashed and dotted lines in panel B refer to private employment shares in NBS and CLSY data, respectively (see Figure 5).

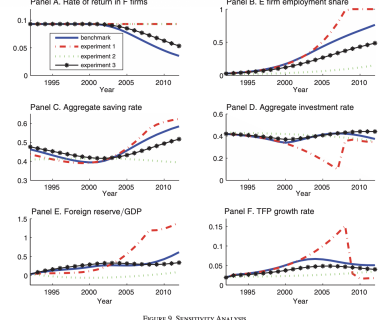


FIGURE 9. SENSITIVITY ANALYSIS.
Notes: The figure shows the evolution of key variables in the calibrated economy (solid line) and various alternative parameterizations. Experiment 1 has a constant ξ . Experiment 2 has $\eta = 0$. Experiment 3 has logarithmic preferences.

7 External surplus / foreign reserves in the benchmark model

7.1 Bank balance sheet

Banks finance domestic lending and foreign assets using worker deposits. The paper writes (eq. (12)):

$$K_{F,t} + \frac{\eta \rho E}{R^l} K_{E,t} + B_t = \zeta^W w_{t-1} N_{t-1}, \quad (17)$$

where ζ^W is the worker saving rate and B_t is net foreign asset holdings.

7.2 Foreign assets and the private employment share (Lemma 3)

Using equilibrium factor prices and the equilibrium mapping between $K_{E,t}$ and $N_{E,t}$, the paper shows:

Lemma 3 (Lemma 3). *During transition,*

$$B_t = \left(\frac{\zeta^W (1 - \alpha) \kappa_F^{\alpha-1}}{(1 + z)(1 + \nu)} - 1 + (1 - \eta) \frac{N_{E,t}}{N_t} \right) \kappa_F A_t N_t \quad (eq. (13)). \quad (18)$$

Intuition for Lemma 3. Deposits scale with last period wages and population. Domestic loan demand scales with the size of the F-sector plus the (limited) fraction of the E-sector that can be financed externally. As the economy reallocates toward E-firms, domestic loan demand does not keep up with deposit growth, so the residual is invested abroad. Hence B_t rises with the E employment share.

8 Post-transition equilibrium (all E-firms)

Once the transition is complete ($N_{E,t} = N_t$), the economy behaves like a standard OLG growth model, and the external surplus eventually disappears. With log utility ($\theta = 1$), the paper derives

the law of motion for capital intensity (eq. (14)):

$$\kappa_{E,t+1} = \frac{\beta}{1+\beta} \frac{\psi}{(1+z)(1+\nu)} \frac{R^l}{R^l - \eta\alpha(1-\psi)\kappa_{E,t}^{\alpha-1}} \kappa_{E,t}^\alpha. \quad (19)$$

Interpretation. After SOEs vanish, the remaining private sector gradually accumulates capital in a neoclassical way. The “China-like” joint pattern (fast growth + foreign surplus) is a *transition phenomenon* tied to reallocation plus borrowing constraints.

9 Quantitative model: multi-period life cycle and calibration

9.1 Why extend the benchmark?

The two-period OLG model is analytically convenient but too stylized for quantitative fit. The quantitative section adopts an Auerbach–Kotlikoff style multi-period OLG with realistic life-cycle saving and capital depreciation.

9.2 Key extensions (what changes)

- Agents live T periods (work for J periods, then retire).
- Capital depreciates at rate δ .
- The intermediation wedge ξ_t can change over time (“financial development”).
- Entrepreneurs’ borrowing is governed by the same limited-pledgeability logic, but solved along a transition path.

9.3 Calibration snapshot (values used in the paper)

In the benchmark quantitative calibration (annual model), the paper uses the following key parameters (selected):

Parameter	Value	Interpretation / target
α	0.5	Capital share (consistent with Bai–Hsieh–Qian style estimates)
δ	0.1	Depreciation rate
ν	0.03	Urban population growth
θ	2	Intertemporal elasticity of substitution
β	0.997	Discount factor (targets aggregate saving)
R	1.0175	Annual real deposit rate (avg. 1998–2005)
η	0.86	Pledgeability (targets external finance share for private firms)
ψ	0.45	Diversion/manager share (disciplining E-firm delegation)
χ	4.79	TFP advantage of E (targets ROA/return gaps and capital–output ratios)
ξ	0.069	Intermediation wedge in the 1990s (targets returns)
z	0.038	Exogenous TFP growth (targets output growth)

(They also assume time-varying ξ_t after 2000 to match the path of aggregate investment.)

9.4 Fig. 8: Benchmark calibration vs data

How read Fig. 8. The calibrated model tracks:

- the rise in the private employment share (Panel B),
- high aggregate saving and investment rates (Panels C–D),
- the rise in reserves/GDP (Panel E),
- and a sizable portion of TFP growth from reallocation (Panel F).

A key quantitative message: reallocation from F to E can account for a large share of measured TFP growth in the data.

9.5 Fig. 9: Sensitivity analysis

Interpretation. The “China-like” joint behavior depends on the combination of (i) a productive but constrained private sector and (ii) a shrinking but credit-favored state sector. When these ingredients are weakened, the model struggles to generate both high growth and a large foreign surplus at the same time.

10 Two-sector extension: why SOEs concentrate in capital-intensive industries

10.1 CES final good and two production technologies

The paper introduces labor-intensive (l) and capital-intensive (k) intermediate goods aggregated via CES (eq. (15)):

$$Y_t = \left(\varphi (Y_t^k)^{\frac{\sigma-1}{\sigma}} + (Y_t^l)^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}, \quad \sigma > 1, \varphi > 0. \quad (20)$$

Firm technologies (eq. (16)) are:

$$y_{j,t}^J = (A_{j,t}^J)^{1-\alpha} (k_{j,t}^J)^\alpha (n_{j,t}^J)^{1-\alpha}, \quad j = l, \quad (21)$$

$$y_{k,t}^J = (A_{k,t}^J)^{1-\alpha} k_{k,t}^J, \quad (22)$$

with $A_{k,t}^F = aA_t$, $A_{l,t}^F = A_t$ and $A_{k,t}^E = a\chi A_t$, $A_{l,t}^E = \chi A_t$. In this section they simplify by assuming log utility, $\eta = 0$, and $\nu = \xi = z = 0$.

10.2 Demand system and prices

With final good as numeraire, the model implies (eq. (17)):

$$\frac{Y_t^k}{Y_t^l} = \left(\varphi \frac{P_t^l}{P_t^k} \right)^\sigma, \quad (23)$$

and the CES price index condition (eq. (18)):

$$\left(\varphi^\sigma (P_t^k)^{1-\sigma} + (P_t^l)^{1-\sigma} \right)^{\frac{1}{1-\sigma}} = 1. \quad (24)$$

10.3 F-firm conditions

For the labor-intensive sector, the equilibrium wage and capital intensity satisfy (eqs. (19)–(20)):

$$w = P_t^l(1 - \alpha)A_{l,t}^F(\kappa_{F,t}^l)^\alpha, \quad (25)$$

$$\kappa_{F,t}^l = \left(\frac{P_t^l \alpha}{R} \right)^{\frac{1}{1-\alpha}}. \quad (26)$$

For the capital-intensive sector under perfect competition, price equals unit cost (eq. (21)):

$$P_t^k(A_{k,t}^F)^{1-\alpha} = R. \quad (27)$$

10.4 Comparative advantage logic and stages of transition (Lemma 4 + Table 2)

Lemma 4 (Lemma 4). *(i) If both sectors have F-firms, E-firms invest only in the labor-intensive sector.*

(ii) If E-firms invest in both sectors, then F-firms invest only in the capital-intensive sector.

Intuition. E-firms are financially constrained and (endogenously) more labor intensive, so they have comparative advantage in labor-intensive production. F-firms are capital intensive (access to finance) and thus concentrate in capital-intensive industries.

Table 2: Investment patterns across transition stages in the two-sector model (paper’s Table 2)

	Stage 1	Stage 2	Stage 3	Stage 4
E-firm invests?				
Labor-intensive	Yes	Yes	Yes	Yes
Capital-intensive	No	No	Yes	Yes
F-firm invests?				
Labor-intensive	Yes	No	No	No
Capital-intensive	Yes	Yes	Yes	No

Interpretation . The private sector (E) enters the labor-intensive sector first. SOEs (F) exit the labor-intensive sector as the private sector accumulates wealth and expands. Only later, once the private sector becomes sufficiently large/wealthy, does it enter the capital-intensive sector. Eventually, in the long run, SOEs are fully displaced.

11 Market power extension: why SOE profits can rise

To reconcile rising SOE profitability with shrinking SOE employment, the paper considers a variant where the F-firm is a *monopolist* in the capital-intensive sector.

Proposition 1 (Proposition 2). *Suppose the capital-intensive sector is monopolized by the F-firm. If the transition starts in the labor-intensive sector (Stage 1 $\rightarrow$ Stage 2), then the monopolist’s markup and profit in the capital-intensive sector increase during Stage 2.*

The optimal monopoly condition is given in the paper as (eq. (22)):

$$\sigma \left(\frac{P_t^k}{R} \right)^{-1} - (\sigma - 1) = \left(1 - \left(\frac{P_t^k}{R} \right)^{-1} \right) \left(\sigma - 1 + \frac{1}{1 - \alpha} \frac{Y_t^F}{Y_t^l} \right) \frac{\varphi^\sigma R^{1-\sigma} (P_t^k/R)^{1-\sigma}}{1 - \varphi^\sigma R^{1-\sigma} (P_t^k/R)^{1-\sigma}}. \quad (28)$$

Intuition for Proposition 1. As SOEs retreat from the labor-intensive sector, the economy becomes more efficient there (more E production), changing relative demand and allowing the capital-intensive monopolist to raise markups. This generates rising SOE profits even as SOE employment declines, aligning the model with Fig. 3.

12 Main takeaways

- The paper provides a unified transition story for **growth + reallocation + external surplus** in China.
- The key wedge is **financial frictions**: productive private firms are constrained, while less productive state firms receive credit.
- Reallocation from F to E drives **TFP growth** and can quantitatively explain a large share of measured TFP growth.
- The same frictions imply domestic savings are not fully absorbed by domestic investment $\Rightarrow$ **reserve accumulation / current account surplus**.
- A two-sector + market power extension helps reconcile **rising SOE profitability** with shrinking SOE employment: SOEs retreat to capital-intensive sectors where they can earn rents.

13 Conclusion

13.1 Summary

Song, Storesletten, and Zilibotti (2011) build a tractable transition model designed to jointly explain four hallmark features of modern China: *(i)* fast output growth, *(ii)* persistently high returns to capital, *(iii)* large within-manufacturing reallocation from state/collective firms toward private entrepreneurial firms, and *(iv)* a large and rising foreign surplus (reserve accumulation / net foreign asset accumulation).

The model combines a standard neoclassical growth backbone with **asymmetric frictions across firm types**. There are two sectors of firms:

- **F-firms (financially integrated, SOE-like)**: they have good access to formal credit (so their marginal product of capital is disciplined by the loan rate), but they operate with *lower productivity* (capturing weaker governance / inability to implement high-productivity delegation).
- **E-firms (entrepreneurial, private-like)**: they can use a *more productive technology* via delegation/managerial incentives, but they face **financial imperfections** (limited pledgeability and thus a borrowing constraint). As a consequence, they must expand mainly through **internal savings** (retained earnings / wealth accumulation).

Because E-firms are more productive but constrained, they exhibit higher returns to capital than F-firms, consistent with the empirical return gap across ownership types.

The transition is driven by a **reallocation dynamic**: as entrepreneurs accumulate net worth, E-firms relax their constraint, expand employment, and progressively displace F-firms. Crucially, as the F-sector shrinks, the banking system collects large household deposits but cannot fully reallocate these funds into E-firms because E-firm borrowing is constrained. The residual savings are invested

in foreign assets, generating a **foreign surplus** *during the transition* even while domestic returns remain high.

A calibrated multi-period version matches key Chinese time series (private employment share, saving and investment rates, foreign asset accumulation, and TFP dynamics). A two-sector extension (labor- vs. capital-intensive production) also rationalizes why SOE-like firms are disproportionately present in capital-intensive industries and retreat faster from labor-intensive industries.

13.2 Main conclusions

1. **A single structural mechanism can jointly generate:** high growth, high investment, high saving, and a growing foreign surplus. The core is the interaction of *credit constraints on high-productivity private firms* with the *downsizing of low-productivity but credit-favored firms*.
2. **Reallocation is a first-order source of measured productivity growth.** In the calibrated transition (1998–2005), aggregate TFP growth is around 5.9% annually in the model, and the paper attributes about 4.2 percentage points (roughly 70%) to *reallocation from less efficient F-firms to more efficient E-firms* rather than to exogenous technology growth alone.
3. **The foreign surplus is a structural outcome of domestic imperfections, not a purely nominal story.** The paper emphasizes that persistent net foreign asset accumulation can arise because constrained high-productivity firms cannot absorb domestic saving quickly enough, so domestic saving is pushed into foreign assets even when domestic returns are high.
4. **Financial development matters quantitatively.** Allowing a gradual decline in the intermediation wedge (a reduced credit/loan friction over time) is important for matching sustained high investment during the transition while the economy reallocates toward constrained firms.
5. **Industry composition patterns are consistent with comparative advantage induced by credit policy.** In the two-sector extension, credit-favored firms have a comparative advantage in capital-intensive activities, while constrained entrepreneurial firms initially specialize in labor-intensive activities and only later expand into capital-intensive production as they accumulate internal funds.

13.3 Intuition

The logic can be summarized as a tight causal chain:

1. **Asymmetric productivity vs. finance:** entrepreneurial firms are technologically superior (higher effective productivity), but cannot freely scale because only a limited fraction of cash flows is pledgeable, so their capital choice is constrained by net worth.
2. **High returns in the constrained sector:** because E-firms are constrained, the marginal product of capital in E-firms stays high; this rationalizes the observed return gap (private > state).
3. **Self-financing generates fast growth of the private sector:** entrepreneurs save aggressively because wealth relaxes the constraint and allows expansion; retained earnings translate into capital deepening and employment expansion in E-firms.

4. **Reallocation raises aggregate TFP:** as E-firms expand and displace F-firms, factor inputs move toward higher-productivity uses, creating a large Solow-residual (TFP) boost *even without assuming extraordinary exogenous technology shocks*.
5. **Why a foreign surplus emerges:** during the transition, household deposits are large, but constrained E-firms cannot borrow enough to absorb all domestic saving. Meanwhile, F-firm activity shrinks. The banking system therefore allocates the residual funds into foreign assets. This generates a rising net foreign asset position (or reserves), *coexisting* with high domestic returns and high domestic investment.
6. **Why the pattern persists for a long time:** constraints relax only gradually through wealth accumulation, so the transition can last many years, delivering a prolonged period of high growth and a persistent external surplus.

13.4 Scope conditions and limitations (brief, for completeness)

- The model treats underlying technology growth largely as exogenous; it does not build a full theory of endogenous innovation/technology adoption.
- It does not aim to fully microfound all determinants of household saving; instead, it highlights the *entrepreneurial saving-constraint* channel and notes that richer saving motives could be integrated.
- Political economy of directed credit and detailed institutional change are not modeled explicitly; these are summarized into reduced-form wedges/constraints.

13.5 Takeaway

China can “grow like China” because high-productivity private firms are credit constrained and therefore expand through internal saving, driving reallocation-driven TFP growth, while the downsizing of credit-favored low-productivity firms pushes excess domestic saving into foreign assets, generating a persistent foreign surplus.